Algorithm 2 Detailed fused Triton dispersion kernel (CR–MLE or MAP)

Require: counts $\mathbf{Y} \in \mathbb{N}_0^{G\times S}$; fitted means $\boldsymbol{\mu} \in \mathbb{R}_+^{G\times S}$; design $\mathbf{X} \in \mathbb{R}^{S\times P}$, $P \in \{2,4\}$; initial dispersions $\boldsymbol{\alpha}^{(0)} \in \mathbb{R}_+^G$; compile-time $h \equiv$ HasPrior; prior means $\mathbf{m}$ and variance σ^2 when $h = 1$; initial step bound κ_0.
Ensure: optimized log dispersions $\mathbf{a}$ and iteration counts $\mathbf{c}$.

parallel for Triton program $g \in \{0, \dots, G-1\}$ **do** (one program per gene)
  Load $\mathbf{y}_g$, $\boldsymbol{\mu}_g$, and $\mathbf{X}$ once into program-local state.
  Set $a \leftarrow \mathrm{clip}\Big(\log \alpha_g^{(0)}, -30, 10\Big)$, $\kappa \leftarrow \kappa_0$, $n_{\mathrm{acc}} \leftarrow 0$, and $c \leftarrow 0$.
  Set $(\ell, q) \leftarrow \mathrm{ObjectiveGradient}(a, \mathbf{y}_g, \boldsymbol{\mu}_g, \mathbf{X}, h, m_g, \sigma^2)$.
  while $c < \mathit{maxit}$ and not converged **do**
    $c \leftarrow c + 1$ and propose $\widetilde{a} \leftarrow a + \kappa q$.
    If $q \neq 0$ and $\widetilde{a} \notin [-30, 10]$, shorten κ so that $\widetilde{a}$ lies on the violated boundary.
    Set $(\widetilde{\ell}, \_) \leftarrow \mathrm{ObjectiveGradient}(\widetilde{a}, \mathbf{y}_g, \boldsymbol{\mu}_g, \mathbf{X}, h, m_g, \sigma^2)$.
    Set $r \leftarrow [\, \widetilde{\ell} \geq \ell + \varepsilon \kappa q^2 \,]$. (Armijo acceptance flag)
    Select $a' \leftarrow r \,?\, \widetilde{a} : a$, $\ell' \leftarrow r \,?\, \widetilde{\ell} : \ell$, and $\Delta \leftarrow \ell' - \ell$.
    Set $(\_, q') \leftarrow \mathrm{ObjectiveGradient}(a', \mathbf{y}_g, \boldsymbol{\mu}_g, \mathbf{X}, h, m_g, \sigma^2)$ and $q \leftarrow r \,?\, q' : q$.
    Set $n'_{\mathrm{acc}} \leftarrow n_{\mathrm{acc}} + r$ and $\kappa_{\mathrm{acc}} \leftarrow \min(1.1\kappa, \kappa_0)$.
    If r and $n'_{\mathrm{acc}} \bmod 5 = 0$, set $\kappa_{\mathrm{acc}} \leftarrow \kappa_{\mathrm{acc}}/2$.
    Set $\kappa \leftarrow r \,?\, \kappa_{\mathrm{acc}} : \kappa/2$, $n_{\mathrm{acc}} \leftarrow n'_{\mathrm{acc}}$, $a \leftarrow a'$, and $\ell \leftarrow \ell'$.
    Converge if $r \wedge (\Delta < 10^{-6} \;\vee\; a < \log(\mathit{minDisp}/10))$.
  end while
  Store $a_g \leftarrow a$ and $c_g \leftarrow c$ in global memory.
end parallel for
function ObjectiveGradient$(a, \mathbf{y}, \boldsymbol{\mu}, \mathbf{X}, h, m, \sigma^2)$
  Set $\alpha \leftarrow e^a$, $\mathbf{W} \leftarrow \mathrm{diag}(\boldsymbol{\mu}/(1 + \alpha\boldsymbol{\mu}))$, and $\mathbf{B} \leftarrow \mathbf{X}^\mathsf{T}\mathbf{W}\mathbf{X}$.
  Compute $\ell \leftarrow \ell_{\mathrm{NB}}(\mathbf{y}; \boldsymbol{\mu}, \alpha) - \frac{1}{2}\log\det(\mathbf{B})$.
  Compute $q \leftarrow \partial\ell/\partial a$, including $\mathrm{tr}\big(\mathbf{B}^{-1}\partial\mathbf{B}/\partial\alpha\big)$ via an unrolled 2×2 formula or 4×4 LU solve.
  if $h = 1$ **then** $\ell \leftarrow \ell - (a - m)^2/(2\sigma^2)$ and $q \leftarrow q - (a - m)/\sigma^2$. (MAP only)
  return (ℓ, q).
end function